

Miles Reid NOTE

for and by Gilbert Allister (10123021)

1. MOTIVATION

This one is reference to Miles Reid Notes, for me to easily read and memorized in my mind. $\mathcal{O}$

2. AFFINE VARIETIES

2.1 NOETHERIAN RINGS

Definition 2.1 (Noetherian). A ring R is called Noetherian if every ideal of R is finitely generated.

Theorem 2.2 (Hilbert Basis Theorem). *Let R be a Noetherian ring. Then the polynomial ring $R[x]$ is also Noetherian.*

3. ZARISKI TOPOLOGY

3.1 ALGEBRAIC SET

3.2 THE CORRESPONDING OF ALGEBRAIC SET

3.3 IRREDUCIBLE ALGEBRAIC SET

4. NULLSTELLENSATZ

4.1 RADICAL IDEALS

4.2 NULLSTELLENSATZ THEOREM

Or also know as Hilbert's zeros theorem.

4.3 COROLLARY OF NULLSTELLENSATZ